

Cross-Cultural Ingredient Networks: A Co-occurrence Analysis of Asian and Western Cuisines

Eric (TianXing) Chen¹ and Serena Chen²

Student number:47368527, 68696426

¹ University of British Columbia Okanagan (UBCO), Department of Computer Science, Canada

² University of British Columbia Okanagan (UBCO), Department of Computer Science, Canada

Abstract. Understanding how different culinary traditions organize and combine ingredients gives us a view of identity and global food practices. In this study we build ingredient co-occurrence networks from an available Kaggle dataset to look at patterns in Asian cuisine and Western cuisine. In the ingredient co-occurrence networks each node is an ingredient and each weighted edge shows that the ingredient co-occurs with another ingredient in a recipe. First we compare the two cuisine groups by looking at network size, network density and centrality distributions. Then we point out core ingredients that act as backbones, in each culinary system. We first put the two data sets together into one network. We then use betweenness centrality to find bridge ingredients. Bridge ingredients link the part of the network and the Western part of the network. Bridge ingredients have a role in mixing Asian and Western cooking. Next we run the Louvain algorithm to find ingredient communities. Ingredient communities group ingredients. Ingredient communities form flavor clusters or new mixed groups. The findings show how network science can show cultural patterns in everyday cooking. The findings also point out the ingredients that help cooking exchange, between Western traditions. The result is clear.

Keywords: Ingredient Networks, Co-occurrence Analysis, Cuisine Comparison, Network Centrality, Food Science

1 Introduction

Food reflects culture. Food reflects geography, history and collective identity.

Each cuisine grows from the ingredients that are found in the region from the cooking practices that people share and, from the exchange that has lasted many years. Asian cuisine and Western cuisine each follow their history. Global trade, migration and fusion cooking now bring cuisine and Western cuisine together. Food scholars describe these interactions in words. Computer methods, network science give a simple way to look at how the ingredients are mixed and how the culinary patterns differ in different cultures.

Network based approaches work well for analyzing the food systems. Network based approaches let us see how the food systems operate.

We have read that the Flavor Network by Ahn et al. (2011) Shows that ingredients can be treated as nodes that share flavor compounds. The network shows patterns in how recipes are built. Later studies built on this idea. Later studies looked at how ingredients appear in recipes. Later studies created recommendation tools. Later studies examined cuisines one by one. Ingredient networks capture cultural information about food. Earlier studies look at only one regional cuisine or only, at worldwide patterns. Earlier studies do not compare the major culinary groups. We notice that the structural differences, between ingredient networks and Western ingredient networks are not studied enough. Furthermore, the mechanisms which link ingredient networks and Western ingredient networks are not studied enough.

We see this gap, as important. Asian cuisine and Western cuisine are two of the world's influential culinary systems. Asian cuisines, South and East Asian traditions use many spices, layers of smell and many ways to cook. Western cuisines use herbs, dairy and baking ingredients. The two cuisines show ways things appear together and different taste patterns. We see that looking at the ingredient network can show how Asian cuisines differ from cuisines. We think the ingredient network can also give clues about culture and how cooking changes, over time. We notice that finding ingredients that show up in both cuisines and Western cuisines can show how mixing of cultures happens in daily cooking.

The study will use network science. The study will compare the structures of Asian and Western ingredient co- networks. The study will identify ingredients that act as bridges, between Asian and Western ingredient co- networks.

Using an available Kaggle dataset of recipes we construct separate ingredient co-occurrence networks for the Asian cuisine for the Western cuisine and for the global network. We examine the network size, the density and the centrality measures to describe the properties of each cuisine. We then identify the core ingredients—nodes, with degree or high eigenvector centrality that serve as the backbones of the ingredient co-occurrence networks. The core ingredients are the hubs. Next we use centrality to detect the bridge ingredients that connect the Asian-dominated region of the global network to the Western-dominated region of the global network. The bridge ingredients link the two sides. Finally we use the Louvain community detection method to find the ingredient communities. The ingredient communities can be flavor clusters or emerging fusion groups.

Through these analyses, we aim to answer four research questions:

Which ingredients are the most central in the global network, what does this reveal in cooking?

How do Asian and Western cuisines differ in network structure?

Which ingredients act as bridges that connect the two culinary systems?

What ingredient communities emerge within the global network, and how do they reflect cultural or fusion patterns?

The study looks at these questions. The study provides a network science view of cooking habits from cultures and provides a data based look, at how Asian cuisines and Western cuisines are different but linked.

2 Related Work

Research that uses network science to study food systems shows that ingredients can be modeled as a network. One known example is the Flavor Network, by Ahn et al. (2011). In that work ingredients are treated as nodes and shared flavor compounds form the edges. Their results reveal global patterns in how ingredients combine in recipes. The study also shows that network-based models can capture cultural rules about how flavors are constructed.

Other studies focus on ingredient co-occurrence instead of shared flavor compounds. Teng et al. (2012) Looked at how ingredients appear in recipes and built ingredient co-networks. The ingredient co-occurrence networks supported recipe recommendation tools. Adaji et al. (2018) Also used recipe networks. Showed that the recipe networks can support decision making systems. The decision-making systems can give personal recipe suggestions. These studies demonstrate that ingredient co-occurrence reflects cooking behavior and can be used to build practical applications.

Network science has also been used to analyze specific regional cuisines. Zhu et al. (2013) studied Chinese regional cuisines and showed that food pairing patterns vary across regions. Jain et al. (2015) analyzed Indian cuisine using flavor networks and found strong anti-pairing behavior, where in anti-pairing behavior ingredients that taste different often sit together. Zhu et al. (2013). Jain et al. (2015) Show that network analysis helps understand the structure of a region's cuisine.

However, earlier work rarely compares major culinary groups directly. I notice most studies focus on a single cuisine or on global patterns, but they do not place Asian and Western cuisines side by side. Few studies examine the ingredients that link the two systems. The idea of “bridge ingredients”—ingredients that appear in both cuisine groups and act as connectors—has not been explored in depth. Another gap in the literature is the lack of structural comparison using measures such as network size, density, and centrality.

This project builds on these ideas and fills the gap. It compares Asian and Western cuisines directly by constructing ingredient co-occurrence networks for both groups. It identifies the core ingredients that serve as backbones in each network and finds the ingredients that act as bridges between the two systems. By applying community detection, this study also highlights how ingredients cluster into flavor groups, including possible fusion clusters.

3 Methodology

This study uses a Kaggle dataset of food recipes that includes the cuisine type and the list of ingredients for each recipe. We want to build ingredient co-occurrence networks for Asian cuisine, Western cuisine, and a combined global set. The steps described

below explain how the data was cleaned, how the networks were created, and how the network measures were computed.

We first cleaned the dataset by converting all cuisine labels and ingredient names to lower case. We removed extra spaces, quotation marks, and brackets from the ingredient lists. Each recipe originally contained the ingredient list as a single string. We split the string at each “|” symbol to produce a clean list of individual ingredients. Each recipe was assigned a unique recipe ID.

We then classified each recipe into one of three cuisine groups. Asian recipes were identified using keywords that matched Indian, Chinese, Japanese, Korean, Thai, Vietnamese, Indonesian, Malaysian, and related regional cuisines. Western recipes were identified using keywords such as American, Mexican, Italian, French, Greek, British, Mediterranean, and Canadian. For those that did not match either list were labeled as “Other.” This classification step produced 5,560 Asian recipes, 1,863 Western recipes, and 586 recipes in the “Other” category.

After classification, we transformed the dataset into a long format. Each row in the long format represents a pair consisting of a recipe ID and one ingredient. Therefore, we could track which ingredients appear together in the same recipe. We removed empty ingredients and duplicate ingredients within the same recipe as well.

To build the ingredient co-occurrence networks, we created all unordered ingredient pairs that appear in the same recipe. Each pair of ingredients forms an edge, and the weight of the edge counts how many recipes contain that pair. We built three networks: an Asian network, a Western network, and a global network that combines both cuisine groups. In each network, the nodes are ingredients and the edges represent co-occurrence.

We then computed several network measures. For each network we calculated the number of nodes, the number of edges, the average degree, and the density. These measures give an overview of the network structure. For every ingredient we computed the degree, the betweenness centrality, and the eigenvector centrality. Degree measures how many neighbors an ingredient has. Betweenness shows how often an ingredient lies on the shortest path between other ingredients. Eigenvector centrality identifies ingredients that connect to other important ingredients.

To identify ingredients that link Asian and Western cuisines, we looked for ingredients that appear in both the Asian and Western recipe sets. We then ranked these ingredients by their betweenness centrality in the global network. Ingredients with high betweenness act as bridges that connect the Asian-dominated region of the network with the Western-dominated region.

Finally, we applied the Louvain community detection algorithm to the global network. This method finds groups of ingredients that co-occur frequently with each other. These communities often reflect flavor clusters or groups of ingredients that appear together in certain types of dishes. Some communities represent traditional cuisine groups, while others may show signs of fusion between culinary systems.

Project Repository: https://github.com/SerenaChen7/COSC421_2025W1

4 Results

This section presents the results of the ingredient co-occurrence networks for Asian cuisine, Western cuisine, and the combined global set. The results include basic network statistics, centrality rankings, bridge ingredients, and community detection outcomes.

4.1 Network Statistics

The Asian network contains 1,431 nodes and 43,418 edges. The Western network contains 904 nodes and 22,769 edges. The global network combines both cuisine groups and contains 1,796 nodes and 61,657 edges.

Average degree differs across the three networks. The Asian network has an average degree of 60.68, while the Western network has an average degree of 50.37. The global network shows the highest value at 68.66. Density values follow the opposite pattern. The Western network is the densest (0.0558), followed by the Asian network (0.0424), while the global network is the sparsest (0.0383). These values show that Western ingredients are more tightly interconnected, while the Asian network is larger and more diverse.

4.2 Central Ingredients

Centrality measures were computed for each network.

Asian Ingredient Network (Top 200, Degree-Colored)

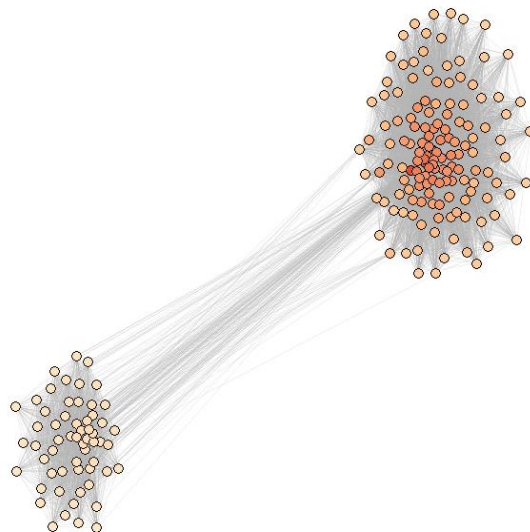


Fig. 1. Asian ingredient co-occurrence subnetwork consisting of the top 200 highest-degree ingredients. (Node color represents degree centrality (darker = higher degree)).

The node colors show degree centrality values through darker nodes which indicate higher degree centrality, and the network consists of two densely connected clusters which share multiple ingredients between them. In the Asian network, the top ingredients by degree include salt, sunflower oil, coriander leaves, garlic, ginger, turmeric powder, onion, red chilli powder, green chillies, and water. Many of these ingredients are staples of Indian and Southeast Asian cooking.

Western Ingredient Network (Top 200, Degree-Colored)

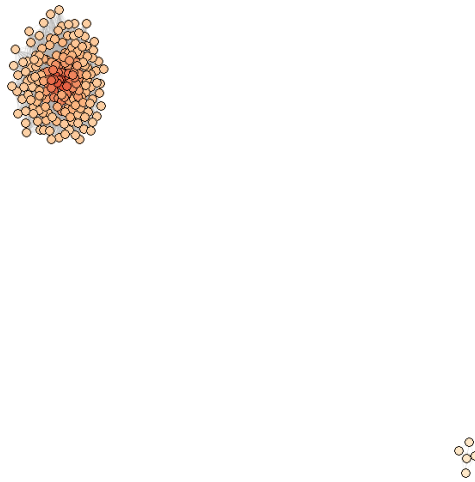


Fig. 2. Western ingredient co-occurrence network (top 200 nodes).

In the Western network, the subgraph shows a single dense core with a small detached cluster because it represents a centralized ingredient structure which is characteristic of Western cuisine. The color scheme indicates degree centrality through darker node colors which represent higher degree values. The highest-degree ingredients include salt, extra virgin olive oil, lemon juice, garlic, sugar, onion, black pepper powder, milk, all-purpose flour, and salt and pepper. These ingredients reflect Western flavor bases and baking practices.

Global Ingredient Network (Top 300, Louvain Communities)

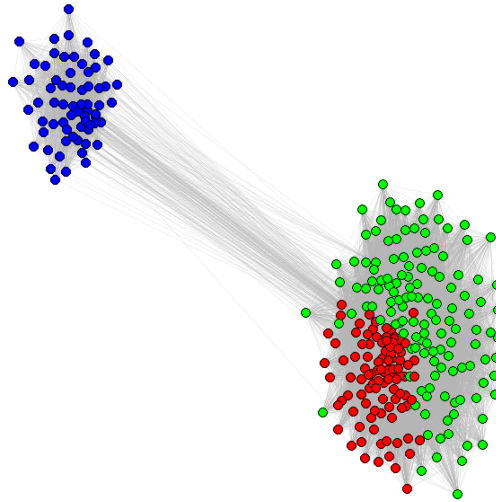


Fig. 3. Global ingredient co-occurrence network showing the top 300 highest-degree ingredients, colored by Louvain communities.

In the global network, the most central ingredients combine patterns from both groups. The figure above shows two distinct clusters which consist of a Western-dominated community (blue) and an Asian-dominated community (green/red). The bridge ingredients which appear in both cuisine groups form cross-cluster edges that connect these regions. The Asian community displays nodes with red and green colors which shows that new fusion patterns are developing. The highest-degree ingredients include salt, garlic, sunflower oil, onion, coriander leaves, sugar, water, ginger, lemon juice, and turmeric powder. These ingredients appear frequently across multiple cuisines and form the backbone of the global network.

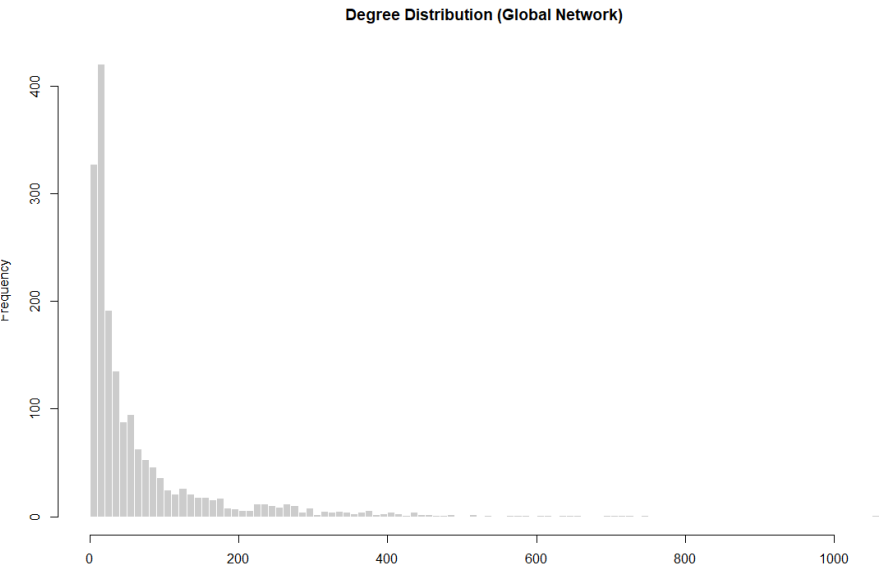


Fig. 4. Degree distribution of the global ingredient co-occurrence network.

The distribution is heavily right-skewed, indicating that most ingredients have very low degree, while a small number of highly common ingredients act as hubs.

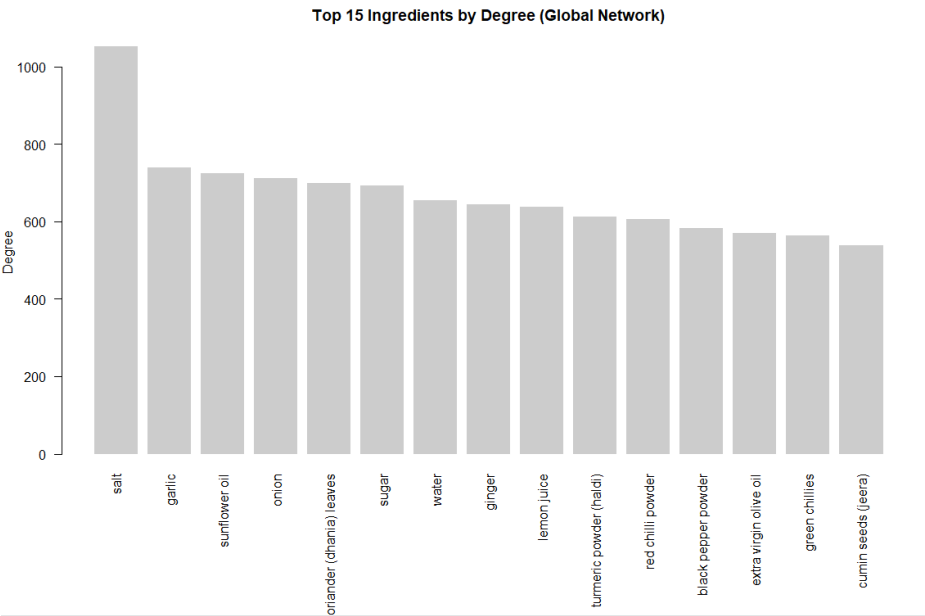


Fig. 5. Top 15 ingredients ranked by degree in the global co-occurrence network.

These ingredients appear across a wide range of recipes and serve as high-connectivity hubs in the global network.

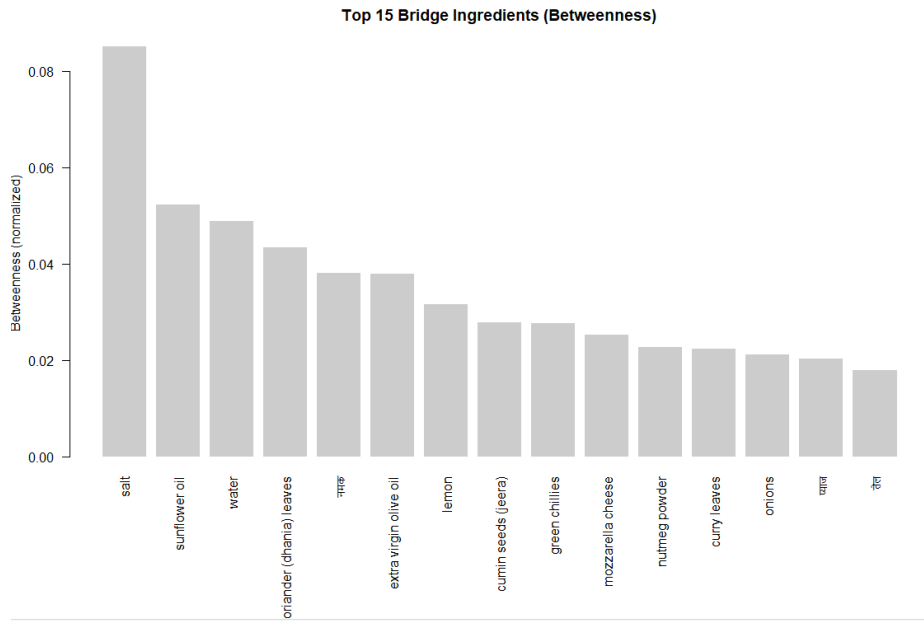


Fig. 6. Top 15 ingredients ranked by betweenness centrality in the global co-occurrence network.

These ingredients act as bridges connecting otherwise separated ingredient communities, enabling ingredient combinations across different cuisines.

4.3 Bridge Ingredients

To identify ingredients that link the two culinary systems, we examined the ingredients that appear in both Asian and Western recipes and ranked them by betweenness in the global network.

Ingredients with the highest betweenness include salt, sunflower oil, water, coriander leaves, extra virgin olive oil, lemon, cumin seeds, green chillies, mozzarella cheese, nutmeg powder, curry leaves, onion, sugar, lemon juice, garlic, and black pepper powder.

The list also contains several ingredients written in Hindi script. These represent ingredients already present in the dataset under English names. Their appearance reflects the large number of Indian recipes in the Asian group.

High-betweenness ingredients often sit between clusters dominated by different cuisine groups. They act as connectors and may contribute to fusion cooking.

4.4 Community Detection

The Louvain algorithm identified ingredient communities within the global network. Some communities contain ingredients commonly used in Indian regional cuisines. Others contain ingredients often used in Western baking or Mediterranean cooking. Several communities mix ingredients from both cuisine groups. These mixed clusters may represent areas where cross-cultural combinations appear more often.

The community structure shows that the global network contains both distinct regional groups and emerging fusion patterns. Ingredients tend to cluster with items they frequently accompany, forming natural flavor groups.

5 Discussion

The results show clear structural differences between Asian and Western ingredient networks. The Asian network contains more nodes and edges because the dataset includes many recipes with diverse spices, aromatics, and regional ingredients. The Western network is smaller but denser, which suggests that Western recipes reuse a more limited set of ingredients in many different combinations. These patterns reflect well-known differences in cooking traditions: Asian cuisines often rely on large spice sets and layered flavor building, while Western cuisines use smaller groups of herbs, dairy, and baking ingredients.

The centrality results support this view. Ingredients like salt, garlic, onion, ginger, coriander leaves, and turmeric powder appear as central hubs in the Asian network. In contrast, extra virgin olive oil, lemon juice, sugar, flour, and black pepper powder play similar roles in the Western network. These ingredients form the core of their respective culinary systems. In the global network these two sets of ingredients merge, and items such as salt, garlic, onion, and water rise to the top, which is expected because they are used across many different cuisines.

The bridge-ingredient analysis highlights how the two networks connect. Ingredients with high betweenness, such as salt, sunflower oil, coriander leaves, extra virgin olive oil, lemon, cumin seeds, onion, sugar, ginger, and garlic, appear in both cuisine groups. These ingredients sit between clusters with strong regional identities. Because they appear across cultural boundaries, they are natural points of contact between cooking traditions. Their position in the global network suggests that they support fusion dishes or may help introduce flavors from one cuisine into another. The presence of duplicated Hindi-script ingredients reflects the strong influence of Indian recipes in the Asian group and shows that part of the dataset contains parallel naming.

The community detection results give a broader view of how ingredients form groups.

Some clusters clearly reflect Asian cooking styles, especially Indian-based spice groups. Others reflect Western styles such as baking or Mediterranean patterns. Several communities mix ingredients from both systems. These mixed groups may represent dishes that blend techniques or flavor profiles, or they may capture universal combinations that appear in many kitchens. The community structure as a whole suggests that

ingredients tend to form coherent flavor groups, but the boundary between Asian and Western patterns is not completely rigid.

Overall, the findings indicate that network science is a useful way to compare cuisines and study how culinary systems connect. The networks capture both the differences and the points of overlap between Asian and Western recipes. They also reveal the ingredients that play central roles within a cuisine and those that link cuisines together. These observations help explain how dishes evolve over time and how cultures influence each other through cooking.

6 Challenges and Reflections

The project encountered multiple obstacles which determined both the research scope and analytical direction. The first major obstacle when we struggled to find suitable data that would answer our research questions. All the available public recipe datasets contain either no specific regional identification or they combine different culinary traditions without showing which dishes belong to Asian or Western categories. The lack of distinguishable Asian and Western cuisine groups in the dataset forced us to perform extra filtering operations before starting any analysis.

And data cleaning proved to be extremely difficult to manage. Because the database contained multiple versions of identical ingredients which used different names and spellings and formats and languages including Hindi script entries. The ingredients used in similar dishes appeared with different lists of ingredients between different recipe versions. Analysis required node count and co-occurrence pattern because these inconsistencies affected the network construction process. The Asian collection contained more items than the Western collection which resulted in an unbalanced situation that endangered the global network structure and produced biased results in our comparison analysis.

Furthermore, creating visualizations turned out to be more difficult than we had expected. The network containing 1800+ ingredients produced un-readable dense graphs when we tried to display the complete network. The Louvain method for community detection required multiple tests because it produced different network partitions when the ingredient set contained noise and the network size varied.

Our research team would use multiple approaches to improve study results when conducting this research again. We would develop an automated system to merge ingredient synonyms through either lexical databases or vector space similarity calculations. The analysis needs ingredient quantity information to create weighted networks which show the importance of ingredients in recipes. It would gain from using different modeling techniques which include flavor compound networks and neural embedding models to reveal additional data points. Maybe the analysis of multiple datasets would help us confirm our findings by using different data sources which would reduce the effect of specific dataset restrictions.

7 Conclusion

This study examined Asian and Western cuisines using ingredient co-occurrence networks built from a Kaggle recipe dataset. By creating separate networks for each cuisine group and a combined global network, we were able to compare their structures and identify ingredients that play important roles in each system. The results show that Asian and Western cuisines differ in the size and density of their networks. Asian recipes use a wider range of ingredients, while Western recipes show tighter connections among a smaller set of items.

Centrality measures highlight the core ingredients within each cuisine. Asian cuisine relies heavily on spices, aromatics, and layered flavor bases, while Western cuisine centers on oils, herbs, dairy, and baking ingredients. In the global network, these patterns blend together, and common items such as salt, garlic, onion, and water appear as key hubs that support many types of dishes.

Bridge-ingredient analysis revealed ingredients that link the two culinary systems. These ingredients sit between clusters associated with each cuisine group and may help explain how fusion dishes emerge. They show how everyday cooking can mix ideas from different cultures. Community detection further showed that ingredients form clear groups, some of which reflect traditional styles, while others show early signs of blending across cuisines.

There are limitations to this work. The dataset contains many Indian recipes, which influences the Asian network. Some ingredients appear under multiple names or scripts, and the co-occurrence approach does not include quantities or cooking methods. These factors may affect the structure of the networks. Future work could merge ingredient synonyms, use weighted networks based on ingredient amounts, incorporate flavor chemistry, or apply additional community detection methods.

Despite these limitations, the analysis demonstrates that network science provides a useful way to study cooking patterns and cultural connections. The networks highlight both the distinct characteristics of Asian and Western cuisines and the ingredients that bring them together.

References

1. Ahn, Y.-Y., Ahnert, S. E., Bagrow, J. P., & Barabási, A.-L. (2011). Flavor network and the principles of food pairing. *Scientific Reports*, 1(1), 1–7.
2. Teng, C.-Y., Lin, Y.-R., & Adamic, L. A. (2012). Recipe recommendation using ingredient networks. In *Proceedings of the ACM International Conference on Health Informatics (IHI '12)*.
3. Adaji, I., et al. (2018). Personality based recipe recommendation using recipe network graphs. In *Social Computing and Social Media: SCSM 2018*, Springer.
4. Jain, A., Nandi, S., & Singh, A. (2015). Analysis of Indian cuisine using flavor network. *arXiv:1502.03815*. Available at: <https://arxiv.org/abs/1502.03815>

5. **Zhu, L., Yang, X., & Huang, J. (2013).**
Analysis of food pairing in regional cuisines of China. *PLOS ONE*, 8(12), e79508.
6. **Newman, M. E. J. (2010).**
Networks: An Introduction. Oxford University Press.
7. **Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, R. (2008).**
Fast unfolding of communities in large networks. *Journal of Statistical Mechanics*, P10008.
8. **Kaggle. (2023).**
Food Recipes Dataset (food_recipes.csv).
Retrieved from: <https://www.kaggle.com/datasets/sarthak71/food-recipes>